MOBILE AND IoT FORENSICS CASE REPORT

SBT-DF205 — Case 1

Android Forensics: Google Pixel 3, Android 10 — Case Reference SBT-DF205-C1

Trainee Name: FASINU TEMIDAYO LUCKY

Registration Number: 2025/FWSD/11378

Course: SBT-DF205 Mobile and IoT Forensics

Institution: International Cybersecurity and Digital Forensics Academy (ICDFA)

Analysis Workstation: Kali GNU/Linux Rolling, kernel 6.16.8+kali-amd64

Report date: 31 October 2026

1. Case Reference and Scope

Field	Value
Case reference	SBT-DF205-C1
Allegation under examination	Unauthorised communication and sharing of information involving a company-issued Google Pixel 3
Evidence source	Joshua Hickman, Android 10 Pixel 3 dataset, Digital Corpora - "Non-Cellebrite Extraction" folder
Archive	Pixel 3.zip, 4.9 GiB (5,247,820,897 bytes), expands to 8.6 GiB / 64,550 files
Archive SHA-256 (baseline and post-extraction, matched)	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a
Acquisition method (per dataset documentation)	Device rooted via TWRP; logical extraction via Cellebrite UFED; converted to non-Cellebrite format
Examiner	Fasinu Temidayo Lucky, 2025/FWSD/11378
Analysis environment	Kali GNU/Linux Rolling, kernel 6.16.8+kali-amd64; unzip 6.00; sha256sum (GNU coreutils) 9.7
Scope	Examination of the supplied extraction only. No rooting, factory reset or live-account access was performed on this archive.

All timestamps in this report are normalised to UTC. Where a raw value is in local time it is explicitly labelled with its offset. The dataset creator's own activity documentation uses UTC-05:00 (EST); this report follows that convention when converting any locally-expressed time (for example, Android camera filenames, which encode local capture time).

2. Executive Summary

This examination analysed a forensic file-system extraction of a Google Pixel 3 (Android 10, build QQ1A.200105.003, security patch 2020-01-01) supplied as a training dataset. The device carried a single primary user profile, self-named "This Is" (user 0), and a secondary profile "User 2" (user 11) created on 2020-02-14. The device generated substantial communication activity across five channels between 2020-01-29 and 2020-02-14: 24 AOSP SMS messages (22 received, 2 sent), 36 Google Messages records (24 SMS-linked plus 12 RCS-only), 19 WhatsApp messages across 3 chats, 9 TextNow messages, and 14 AOSP call records.

The device owner's identity is corroborated across five independent sources - the user-profile display name, the aggregated AOSP contact "Josh Hickman", the Google account thisisdfir@gmail.com, and matching username/name/email fields in the TextNow application preferences - though none of these is a cryptographic or biometric identifier (Section 7).

Of the 24 SMS messages, 22 are automated two-factor-authentication and notification codes (LINE, Telegram, Facebook, Microsoft, Imgur, Skout, TikTok, Dust, Wickr, Wire). Only one thread (Thread 17, 2 messages) is human-authored free text, and both messages in it were sent (type=2) by the device itself to a number sharing the device's own area code; the second message's content ("Wow, the messages sync over between accounts") reads as a self-test, not a conversation with a third party. A richer, genuinely interactive exchange exists in WhatsApp (self-contact "Josh Hickman") and in TextNow (with +19842032223), the latter containing the explicit statement "Yep! Have to keep generating data" - consistent with deliberately generated test/training content rather than covert real-world communication.

Location evidence places the device in the Raleigh/Holly Springs, North Carolina area on 2019-09-17 (photograph with embedded GPS, corroborated by a nearby Google Maps saved place) and on 2020-02-09 (WhatsApp live-location share, corroborated in time by a camera photo and an SMS message within the same hour). A further Google Maps saved-place entry sits in the New York City area, with no corroborating evidence in this examination.

3. Evidence Acquisition and Integrity (Task 1)

The evidence archive (Pixel 3.zip) was received at 2026-10-27T20:18:00Z local-equivalent and its SHA-256 hash recorded as a baseline before any extraction. A working copy was extracted to a dedicated case directory; the archive itself was never modified. A post-extraction re-hash of the archive confirmed the value was unchanged, and the archive was then retained read-only.

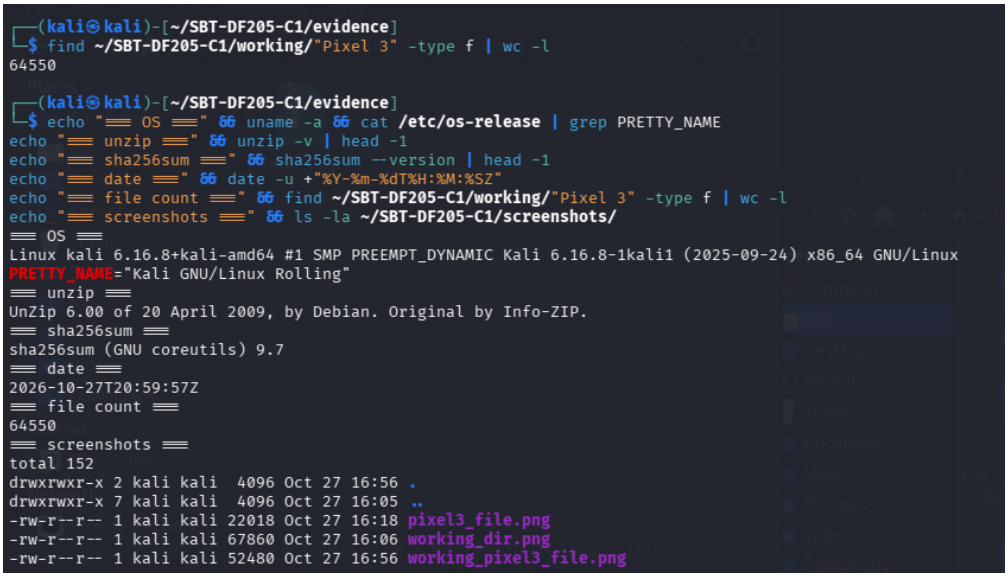


Figure 3.1 - Analysis environment and tool-version verification (Kali Rolling, unzip 6.00, sha256sum 9.7), and the 64,550-file extraction count confirming a complete expansion.

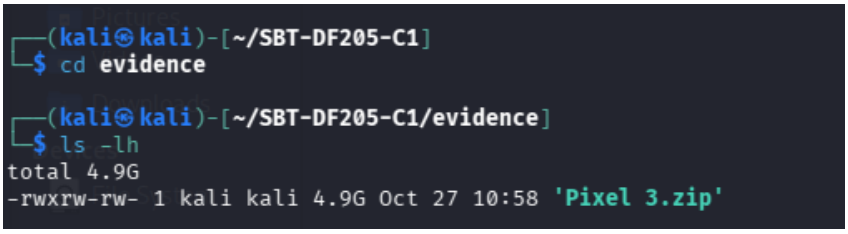


Figure 3.2 - Evidence archive located in the case evidence/ folder, 4.9 GiB.

Step	Result
Baseline SHA-256 (pre-extraction)	ca6918ef...4a8aa00a
Post-extraction SHA-256	ca6918ef...4a8aa00a - identical
Extraction tool	UnZip 6.00 of 20 April 2009, by Info-ZIP
Extracted size / file count	8.6 GiB / 64,550 files
Working-copy policy	Analysis performed only against the extracted working copy; original archive retained unmodified

A per-file evidence register (Appendix A) was maintained for every database and configuration file examined in depth, recording source path,

size, SHA-256 and the UTC time of capture. Database sidecar files (-wal / -shm) were checked for and preserved where present; none were present for the two mmssms.db copies examined in Task 4, which is itself a finding (Section 5.3).

4. Method

The extraction was examined with an archive extractor, sqlite3 (CLI) and DB Browser for SQLite, a text editor, and manual XML/property-file review; ExifTool and a spreadsheet editor were available but EXIF review was limited to what the Photos application's own database exposed (Section 9, Issue I-05). All command output used for a finding was captured contemporaneously to a text file (reports/raw_*.txt) rather than retyped afterwards, and cross-checked where possible against a second, independent query or a DB Browser GUI screenshot - a practice that itself surfaced one discrepancy, documented in Section 9.

Two representations of the Telephony Provider database exist on this device - a Credential-Encrypted (CE) copy under data/data/com.android.providers.telephony/databases/ and a Device-Encrypted (DE) copy under data/user_de/0/com.android.providers.telephony/databases/. Both were hashed and queried; the CE copy (24 SMS / 17 threads / 7 PDU) is used throughout as the authoritative message source, and the DE copy (0 / 0 / 0, same size, different SHA-256) is retained only as corroboration that Android 10 migrates SMS/MMS content to Credential-Encrypted storage.

5. Findings

5.1 Device Identity and User Profiles (Task 2)

Property key	Value	Source file
ro.product.product.manufacturer / .model	Google / Pixel 3	product/build.prop
ro.product.product.device / .name	blueline	product/build.prop
ro.product.build.id	QQ1A.200105.003	product/build.prop
ro.product.build.fingerprint	google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys	product/build.prop
ro.build.version.release / .sdk	10 / 29	product/build.prop

Property key	Value	Source file
ro.build.version.security_patch	2020-01-01	system/build.prop
ro.build.date.utc	1576611252 -> 2019-12-17 19:34:12 UTC (ROM compile time)	system/build.prop



Figure 5.1 - First grep attempt against product/build.prop using the conventional ro.product. keys returned no output (see Section 9, Issue I-01, for the namespace explanation and fix).*



Figure 5.2 - system/build.prop queried successfully for build ID, security patch, build date and build type.

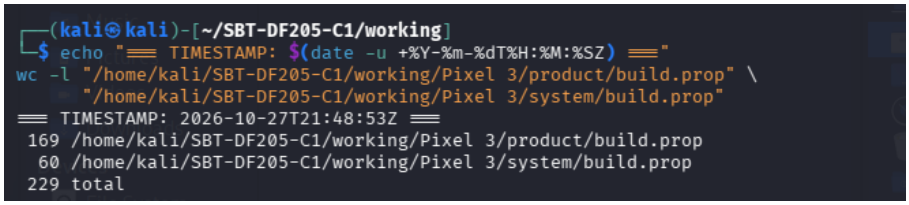


Figure 5.3 - Line-count sanity check (169 + 60 = 229 lines) confirming both build.prop files were read in full.

User profiles were established from data/system/users/. Two profiles exist:

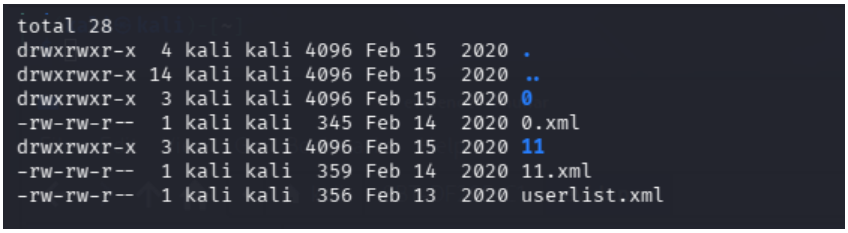


Figure 5.4 - Directory listing of data/system/users/, showing profile folders 0 and 11 alongside userlist.xml.

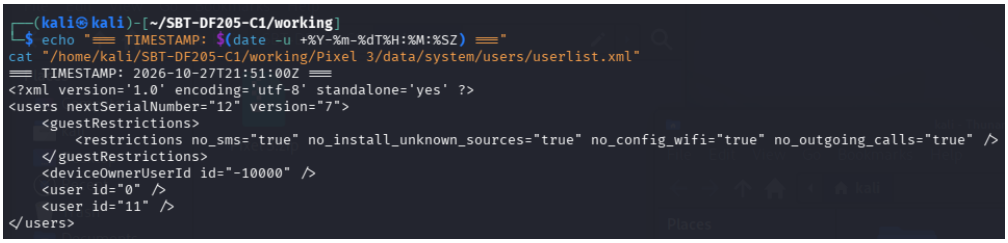


Figure 5.5 - userlist.xml: two users (0, 11), deviceOwnerUserId=-10000, guest restrictions present.

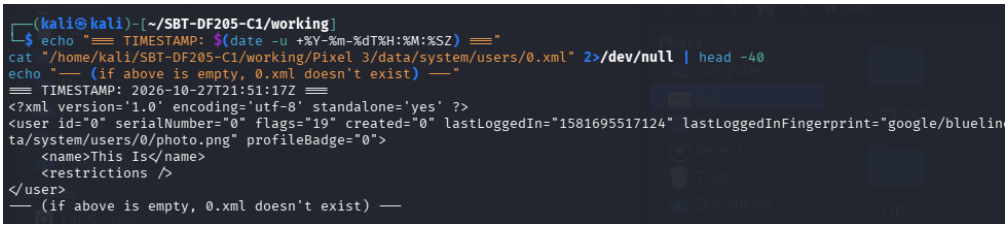


Figure 5.6 - 0.xml: primary profile, display name "This Is", flags=19, lastLoggedIn=1581695517124 ms (2020-02-14 15:51:57 UTC).

User	Display name	Created	Last logged in	Flags
0 (primary)	This Is	0 (epoch - pre-dates meaningful conversion; treated as "at provisioning")	2020-02-14 15:51:57 UTC	19 (primary + admin + initialized)
11 (secondary)	User 2	2020-02-14 01:39:18 UTC	2020-02-14 12:50:01 UTC	16 (initialized)

Corroboration: the AOSP contact aggregation (Section 5.3) resolves ten raw_contacts rows across eight synced accounts to a single aggregate contact, display name "Josh Hickman", whose phone numbers and the Google account thisisdfir@gmail.com recur in TextNow's own preferences (username thisisdfir, name "This Is DFIR", email thisisdfir@gmail.com - Section 5.5). This is an independent second source linking the profile display name "This Is" to the identity cluster "Josh Hickman / thisisdfir".

5.2 Settings and Connectivity (Task 3)

Setting	Key	Value	Source
Bluetooth device name	bluetooth_name	Pixel 3	settings_secure.xml
Bluetooth MAC address	bluetooth_address	7C:D9:5C:AC:A2:CE	settings_secure.xml
Mock-location allowed	mock_location	0 (disabled)	settings_secure.xml
Bluetooth radio state	bluetooth_on	0 (off)	settings_global.xml
Wi-Fi radio state	wifi_on	1 (on)	settings_global.xml
Airplane mode	airplane_mode_on	0 (off)	settings_global.xml
System locale	system_locales	en-US	settings_system.xml

Wi-Fi: WifiConfigStore.xml contains an empty NetworkList but a residual MacAddressMap entry referencing SSID "CcookiesDcastleR5" (WPA_PSK) at BSSID ae:f6:c7:a8:a5:c2. This is saved/residual configuration data only - it is not, by itself, evidence of a proven connection to that network at a specific time, since the active NetworkList is empty and no association timestamp is present in this file. No Bluetooth paired-device list was located in the captured evidence beyond the device's own name/address above.

5.3 SMS, MMS, Contacts, Calls and Calendar (Task 4)

5.3.1 Two copies of mmssms.db

Copy	Path (abbreviated)	Size	SHA-256 (abbrev.)	mtime (UTC)	SMS / threads / PDU
CE (authoritative)	.../data/com.android.providers.telephony/databases/mmssms.db	118,784 B	cd5452f5...3f1342	2020-02-14 15:00:54	24 / 17 / 7
DE (corroboration only)	.../data/user_de/0/.../mmssms.db	118,784 B	d0ce9034...34c827	2020-02-14 04:34:12	0 / 0 / 0

Both files are the same size and share the same schema, but their SHA-256 values differ - confirming different content, not a copy error. The DE copy is functionally empty of user content, consistent with Android 10 migrating Telephony Provider message content to Credential-Encrypted storage. The CE copy's mtime is the later of the two, consistent with the device being

unlocked when the last write occurred. Neither copy has a -wal or -shm sidecar, indicating both were cleanly checkpointed before extraction. All subsequent SMS/MMS findings use the CE copy.

5.3.2 Selected SMS records and Thread 17 reconstruction



11	12	837402	2020-01-29 18:50:13	1	0	FB-55286 is ThisIs's Facebook confirmation code
12	13	32665	2020-01-29 18:50:24	1	0	Confirmed! To edit SMS preferences go to m.facebook.com/sett
13	5	732873	2020-01-29 20:16:10	1	0	<#> Your Imgur verification code is 4907230 WXYiuevlJ2b
14	14	44398	2020-01-29 21:15:33	1	0	369967 is your verification code for Dust - a safer place to
15	15	+17162394257	2020-02-01 01:00:03	1	1	Your Wickr Me confirmation code is 8434776
16	16	+13233208934	2020-02-01 01:04:12	1	1	Your Wire code is 172488. Open https://account.wire.com/v/1
17	13	32665	2020-02-03 19:18:46	1	0	Xavier Mccollum is a new Facebook friend suggestion: https
18	13	32665	2020-02-05 16:22:21	1	0	ThisIs, you have 1 new notification on Facebook: https://f
19	13	32665	2020-02-07 08:15:59	1	0	ThisIs, you have 1 new notification on Facebook: https://f
20	13	32665	2020-02-08 18:17:01	1	0	ThisIs, you have 1 new notification on Facebook: https://f
21	17	9195790479	2020-02-09 18:41:42	2	1	I am on my way to the coffee shop. Do you want anything?
22	13	32665	2020-02-11 18:02:40	1	0	ThisIs, you have 6 new notifications on Facebook: https://
23	13	32665	2020-02-12 19:12:18	1	0	ThisIs, you have 9 new notifications on Facebook: https://
24	13	32665	2020-02-13 21:12:24	1	0	ThisIs, you have 9 new notifications on Facebook: https://
25	17	919-579-0479	2020-02-14 13:09:03	2	1	Wow, the messages sync over between accounts.

Figure 5.7 - DB Browser view of mmssms.db sms table (rows 11-25), used to verify the exact _id/time/body association for each message - see Section 9, Issue I-03, for a discrepancy this screenshot resolved.

SMS _id	Thread	Direction	Address	UTC time	Read	Body (truncated)
2	4	INBOX	+13233205525	2020-01-29 18:20:45	0 (unread)	<#> Please enter 304868 into LINE...
12	13	INBOX	837402	2020-01-29 18:50:13	0 (unread)	FB-55286 is ThisIs's Facebook confirmation code
13	13	INBOX	32665	2020-01-29 18:50:24	0 (unread)	Confirmed! To edit SMS preferences go to m.facebook.com/settings...
21	17	SENT	9195790479	2020-02-09 18:41:42	1 (read)	I am on my way to the coffee shop. Do you want anything?
25	17	SENT	919-579-0479	2020-02-14 13:09:03	1 (read)	Wow, the messages sync over between accounts.

Thread 17 reconstruction: both messages (_id=21 and _id=25) are type=2 (SENT), addressed to the same number in two formats (9195790479 / 919-579-0479), normalised by canonical_addresses._id=17 to +19195790479. There are no incoming (type=1) replies in this thread. The content of the second message explicitly states the device owner was testing cross-

account message sync, not conversing with a third party. This is the only human-authored thread of the 20 in the database; the remaining 19 are automated two-factor-authentication or notification messages.

Statistics (verified by direct query, Queries.sql): 24 total SMS; 22 received (type=1) / 2 sent (type=2); 20 unread / 4 read; 17 of 20 threads carry at least one message (the last three, _id 18-20, are empty). Thread 13 (Facebook notifications, 8 messages, all received from address 32665) is the largest thread.

20 of 24 SMS are unread. This is unusual for an actively used personal device and is more consistent with a device generating or receiving automated test traffic than one in routine daily use - a point carried into the conclusion (Section 8).

5.3.3 Contacts and calls

contacts2.db aggregates ten raw_contacts rows, from eight distinct synced accounts (Google thisisdfir@gmail.com, Telegram, Silent Circle, Signal, Skype, Viber, WhatsApp, Duo), into a single aggregate contact_id=10, display name "Josh Hickman". Two further raw_contacts rows (Telegram-sourced) are marked deleted=1. The per-app identifiers attached to this aggregate (phone +1 919-579-0479 in most entries, WhatsApp JID 19195790479@s.whatsapp.net, Viber number +19195790479) all converge on the same number used in the SMS self-conversation above.

calllog _id	Number	UTC time	Duration	Type	Notes
7 / 8	+19195121037	2020-02-12 15:37:29 / 15:38:21	0s / 3s	MISSED / VOICEMAIL	Geocoded "North Carolina"; voicemail 52s after the missed call
9 / 10	+15755863247	2020-02-13 00:52:44 / 00:53:33	0s / 3s	MISSED / VOICEMAIL	Geocoded "Questa, NM" - a different US region
11 / 12	9195790479	2020-02-14 01:29:01 / 01:30:24	11s / 65s	OUTGOING / INCOMING	Self-call pair, contact "Josh Hickman"
13 / 14	9195790479	2020-02-14 13:05:57 / 14:41:18	86s / 91s	OUTGOING / INCOMING	Second self-call pair

14 calls total: 8 missed, 2 voicemail, 2 incoming, 2 outgoing. Both incoming/outgoing pairs on 2020-02-14 are self-calls to the same contact ("Josh Hickman", 9195790479) also linked through SMS and WhatsApp - a third independent corroboration of that identity cluster, and further

evidence of self-testing behaviour rather than third-party communication for those specific events.

5.3.4 Calendar Provider

calendar.db contains one calendar record but zero rows in every user-content table: Events=0, Attendees=0, Reminders=0, CalendarAlerts=0, ExtendedProperties=0, EventsRawTimes=0, Instances=0. Only static/reference tables are populated (Calendars=1, Colors=35, CalendarCache=4, CalendarMetaData=1). This is a documented absence, not a query failure: the schema and row-count queries both executed successfully and returned zero for every content table.

5.4 Google Messages and Dialer Comparison (Task 5)

bugle_db (Google Messages) contains 36 message rows against mmssms.db's 24 SMS - a difference of 12. Three rows were confirmed to be the same underlying message as their AOSP counterpart via the sms_message_uri column:

bugle_db _id	sms_message_uri	received_utc (bugle)	AOSP body
1	content://sms/2	2020-01-29 18:20:45	<#> Please enter 304868 into LINE...
2	content://sms/3	2020-01-29 18:30:46	Telegram code: 59389...
13	content://sms/12	2020-01-29 18:50:24	Confirmed! To edit SMS preferences...

The remaining 12 bugle_db-only rows are RCS messages (message_protocol=5) not present in AOSP at all, including a separate conversation (conversation _id=15, contact "Josh Hickman", latest_message_id=38, snippet "That is interesting.") that exists only in Google Messages. This demonstrates that Google Messages overlays RCS on top of the AOSP SMS provider: an investigator relying on mmssms.db alone would miss this RCS activity entirely, including its content, which could not be examined further as it falls outside the AOSP schema captured here.

Google Dialer's annotated_call_log.db contains 12 call records against AOSP calllog.db's 14. Two records (calllog._id=7 and _id=13) match exactly on number, timestamp, duration and type, validating the AOSP extraction. The two AOSP VOICEMAIL records (_id=8, _id=10) are absent from the Dialer-annotated log, because Google Dialer does not annotate carrier-voicemail-triggered events the same way as a dialled/received call. Conversely, the Dialer schema carries fields AOSP does not (spam_status, is_voicemail_call, call_screening_details, transcription) - neither source is a complete record alone.

5.5 WhatsApp and TextNow (Task 6)

5.5.1 WhatsApp

WhatsApp's own registration record (registration.RegisterPhone preferences) lists phone number 9195794674, formatted (919) 579-4674. This differs from the AOSP/SIM number used throughout SMS and calls, 9195790479 - the two numbers share the area code and most digits but are not the same number (4674 vs 0479 in the last four digits). This is forensically notable: either a SIM change, dual-SIM use, or a shared/synced WhatsApp account would explain it, and the evidence does not distinguish between these.

wa.db's wa_contacts table holds only 3 rows: status@broadcast (the system broadcast channel), 16505434800@s.whatsapp.net (a number not otherwise seen in this evidence set), and 19195790479@s.whatsapp.net, display name "Josh Hickman" - the same number as the AOSP self-contact. msgstore.db contains 19 messages across 3 chats, 8 media attachments and 2 WhatsApp voice calls.

msgstore_id	UTC time	Direction	Content / event
12	2020-02-08 20:55:14	Sent	"Hi there! I switched over."
13	2020-02-08 20:56:05	Received	"Awesome!"
14	2020-02-08 20:57:05	Received	Image, 21,890 bytes (IMG-20200208-WA0000.jpg)
15	2020-02-08 20:57:49	Sent	"The Osbourne!"
16	2020-02-08 20:58:12	Sent	Image, 87,909 bytes (IMG-20200208-WA0001.jpg)
17	2020-02-09 18:00:14	Sent	Live location "Here I am!!" at (35.6609133, -78.8798813)
18	2020-02-09 18:00:33	Sent	Live-location update at (35.6609423, -78.8798167)
19	2020-02-09 18:05:02	Sent	Static location at (35.6609311, -78.8798382)

All 19 msgstore messages were reviewed: 7 are status@broadcast rows (WhatsApp Status/Stories, not person-to-person messages, and of minimal forensic value here), 3 are a brief media exchange with 16505434800@s.whatsapp.net, and the 9 messages with the self-contact "Josh Hickman" shown above form the only substantive conversation,

including the two location-sharing events used in the correlation in Section 6.

5.5.2 TextNow

TextNow's own preferences give a complete account profile: username thisisdfir, phone 9842352054, email thisisdfir@gmail.com, first/last name "This Is" / "DFIR". The combined name "This Is DFIR" echoes the user-profile display name "This Is" (0.xml) and the email matches the Google account already seen in contacts2.db - a third independent corroboration of the same identity cluster (Section 5.1).

textnow _id	Direction	UTC time	Content
16	Incoming	2020-02-08 20:00:46	"Hi there!"
18	Outgoing	2020-02-08 20:01:27	"Back at it again, I see."
19	Incoming	2020-02-08 20:03:06	"Yep! Have to keep generating data."
20 / 21	Outgoing	2020-02-08 20:05:41	Audio call placed and ended
23	Incoming	2020-02-08 20:08:22	"Missed call from +19842032223" (auto-generated)
24 / 22	Both	2020-02-08 20:09:16 / 20:10:25	Picture messages exchanged
25	Incoming	2020-02-08 20:11:10	"I wonder why I can't call you or send you pictures."

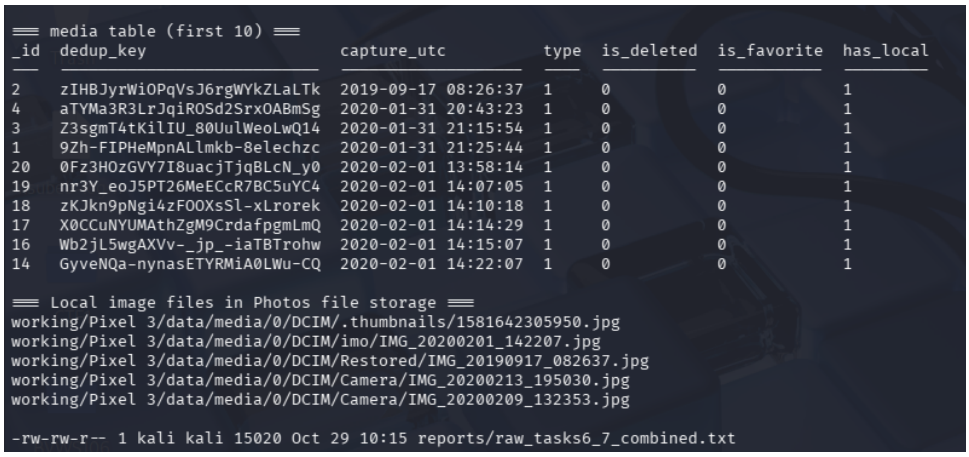
All nine records involve +19842032223 (contact "+1(984) 203-2223"), a US 984-area-code number. The exchange explicitly admits, in _id=19, to "generating data" - direct, first-person evidence that at least this conversation was produced for test/training purposes. +19842032223 also matches TextNow's own LAST_CALL_DESTINATION preference value, correlating the participant with a second source within the same application.

5.6 Maps and Photographs (Task 7)

gmm_sync.db's sync_item_data table (corpus 11 = saved places, per the course slides) holds three geocoded entries:

Latitude / Longitude	Timestamp	Approx. location
35.734602, -78.636654	2019-04-04 19:49:03 UTC	Near Raleigh, NC (within ~100 m of the GPS-tagged photo below)
35.659596, -78.872848	0 (unset in source data)	Near Cary, NC, ~10 km SW of Raleigh
40.784374, -74.065536	0 (unset in source data)	New York City area (Manhattan)

gmm_storage.db's gmm_storage_table holds 4 cached objects: two URI blobs (57,582 and 212 bytes), one CompletedNavigation record (11,623 bytes, indicating at least one completed turn-by-turn navigation was cached on-device), and one CommuteSetupNotificationModel (37 bytes). All four are Protocol Buffer-serialised; without the Google Maps .proto schema, which was not available within this examination's scope, only their presence and size can be confirmed - their content was not decoded.

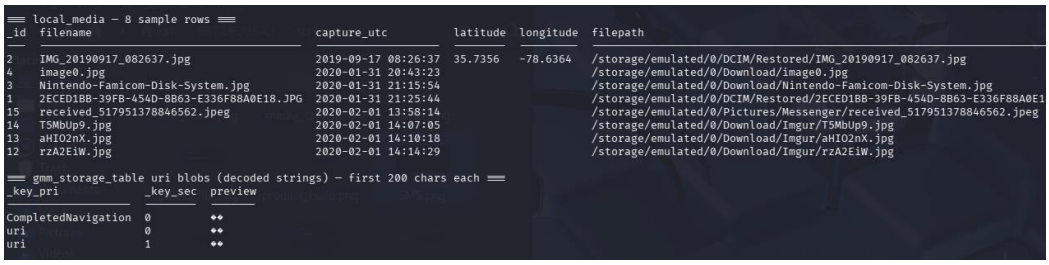


```
═══ media table (first 10) ═══
_id  dedup_key                    capture_utc          type  is_deleted  is_favorite  has_local
───  ──────────────────────────   ──────────────────   ────  ──────────  ───────────  ─────────
2    zIHBJyrWiOPqVsJ6rgWYkZLaLTk  2019-09-17 08:26:37  1     0           0            1
4    aTYMa3R3LrJqiROSd2SrxOABmSg  2020-01-31 20:43:23  1     0           0            1
3    Z3sgmT4tKilIU_80UulWeoLwQ14  2020-01-31 21:15:54  1     0           0            1
1    9Zh-FIPHeMpnALlmkb-8elechzc  2020-01-31 21:25:44  1     0           0            1
20   0Fz3HOzGVY7I8uacjTjqBLcN_y0  2020-02-01 13:58:14  1     0           0            1
19   nr3Y_eoJ5PT26MeECcR7BC5uYC4  2020-02-01 14:07:05  1     0           0            1
18   zKJkn9pNgi4zFOOXsSl-xLrorek  2020-02-01 14:10:18  1     0           0            1
17   X0CCuNYUMAthZgM9CrdafpgmLmQ  2020-02-01 14:14:29  1     0           0            1
16   Wb2jL5wgAXVv-_jp_-iaTBTrohw  2020-02-01 14:15:07  1     0           0            1
14   GyveNQa-nynasETYRMiA0LWu-CQ  2020-02-01 14:22:07  1     0           0            1

═══ Local image files in Photos file storage ═══
working/Pixel 3/data/media/0/DCIM/.thumbnails/1581642305950.jpg
working/Pixel 3/data/media/0/DCIM/imo/IMG_20200201_142207.jpg
working/Pixel 3/data/media/0/DCIM/Restored/IMG_20190917_082637.jpg
working/Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg
working/Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg

-rw-rw-r-- 1 kali kali 15020 Oct 29 10:15 reports/raw_tasks6_7_combined.txt
```

Figure 5.8 - Google Photos media table, 10 most recent entries; all type=1 (photo), has_local=1, is_deleted=0.



```
═══ local_media — 8 sample rows ═══
_id  filename                                  capture_utc          latitude  longitude  filepath
───  ──────────────────────────────────────   ──────────────────   ────────  ─────────  ──────────────────────────────────────────────────────────────────────
2    IMG_20190917_082637.jpg                  2019-09-17 08:26:37  35.7356   -78.6364   /storage/emulated/0/DCIM/Restored/IMG_20190917_082637.jpg
4    image0.jpg                               2020-01-31 20:43:23                       /storage/emulated/0/Download/image0.jpg
3    Nintendo-Famicom-Disk-System.jpg         2020-01-31 21:15:54                       /storage/emulated/0/Download/Nintendo-Famicom-Disk-System.jpg
1    2ECED1BB-39FB-454D-8B63-E336F88A0E18.JPG 2020-01-31 21:25:44                       /storage/emulated/0/DCIM/Restored/2ECED1BB-39FB-454D-8B63-E336F88A0E1
15   received_517951378846562.jpeg            2020-02-01 13:58:14                       /storage/emulated/0/Pictures/Messenger/received_517951378846562.jpeg
14   T5MbUp9.jpg                              2020-02-01 14:07:05                       /storage/emulated/0/Download/Imgur/T5MbUp9.jpg
13   aHIO2nX.jpg                              2020-02-01 14:10:18                       /storage/emulated/0/Download/Imgur/aHIO2nX.jpg
12   rzA2EiW.jpg                              2020-02-01 14:14:29                       /storage/emulated/0/Download/Imgur/rzA2EiW.jpg

═══ gmm_storage_table uri blobs (decoded strings) — first 200 chars each ═══
_key_pri            _key_sec  preview
──────────          ────────  ──────
CompletedNavigation 0         ••
uri                 0         ••
uri                 1         ••
```

Figure 5.9 - local_media table (8 rows retrieved) with EXIF-derived GPS where available, alongside the gmm_storage_table BLOB previews, which could not be decoded as text (Protocol Buffer data).

Of the 8 local_media rows retrieved, only one carries GPS coordinates: _id=2, IMG_20190917_082637.jpg, captured 2019-09-17 08:26:37 UTC at (35.7356, -78.6364) - within approximately 100 m of the Maps saved-place entry above. This is a genuine cross-source corroboration: an independently GPS-tagged photograph sits almost exactly where the user separately saved a place in Google Maps. The remaining 7 local_media rows (all Download/, Messenger/ or Restored/ files from 2020-01-31 and 2020-02-01) carry no latitude/longitude, consistent with GPS tagging being disabled or stripped for those captures.

Comparison attempted between Maps and media evidence (Task 7c): the WhatsApp live-location share on 2020-02-09 (Section 5.5.1, GPS 35.6609, -78.8799) is temporally close to the filename-derived photo time (18:23:53 UTC, unverified) and the SMS "coffee shop" message (18:41:42 UTC, verified) - full chain in Section 6. The strength of this correlation rests entirely on the WhatsApp GPS data, which is a direct, verified device-generated coordinate; the photo and SMS add temporal proximity only, not independent location confirmation. Saved places and photo metadata alone do not prove a person visited a location, and no claim of physical presence is made here beyond what the WhatsApp GPS itself establishes for that specific five-minute window.

6. Timeline and Correlation (Task 8)

A full 21-event timeline spanning six artefact families (build properties, Maps, Photos, SMS, WhatsApp, TextNow, Calls, filesystem metadata, user profile) is provided as Appendix C and as the accompanying Timeline.csv. Three correlations are presented below; duplicate representations of the same underlying message or call (e.g. the same SMS appearing in both mmssms.db and bugle_db) are counted once.

Correlation 1 - Physical presence near Holly Springs/Raleigh, NC on 2020-02-09

UTC time	Event	Source
18:00:14	WhatsApp live location shared to self-contact "Josh Hickman"	msgstore.messages._id=17 (verified GPS)
18:00:33 - 18:05:02	Two further location updates, same immediate area	msgstore.messages._id=18, 19 (verified GPS)
18:23:53 (unverified)	Camera photo IMG_20200209_132353.jpg captured (filename-derived, no DB/GPS record)	File listing only - see Issue I-05

UTC time	Event	Source
18:41:42	SMS sent: "I am on my way to the coffee shop..."	mmssms.db.sms._id=21 (verified)

Strength: the WhatsApp GPS points (18:00-18:05 UTC) are direct, device-recorded coordinates and are strong evidence the device was in that specific area at that time. Limitation: the photo and SMS are temporal correlations only - neither carries its own GPS in this examination, so they corroborate a timeframe, not an independently confirmed location.

Correlation 2 - Owner identity via five independent sources

- data/system/users/0.xml -> display name "This Is"
- contacts2.db.accounts -> thisisdfir@gmail.com
- contacts2.db.raw_contacts (aggregate contact_id=10) -> "Josh Hickman", phone 919-579-0479
- TextNow preferences -> username thisisdfir, name "This Is DFIR", email thisisdfir@gmail.com, phone 984-235-2054
- WhatsApp registration -> phone 919-579-4674

Strength: five independent application databases converge on one identity cluster. Limitation: all are self-entered display names/usernames, not cryptographically verified identity, and the dataset's documented creator is itself named Joshua Hickman - consistent with (though not proof of) this being the dataset creator's own test identity rather than a victim's or suspect's in the ordinary sense.

Correlation 3 - RCS bridge between AOSP SMS and Google Messages

Three SMS rows appear identically in both mmssms.db and bugle_db (matched via sms_message_uri, Section 5.4); bugle_db additionally holds 12 RCS-only messages absent from AOSP entirely. Strength: the URI links are direct and unambiguous. Limitation: the content of the 12 RCS-only messages could not be examined further within this evidence set's AOSP-centred capture.

Conclusion on the allegation

"Unauthorised communication and sharing of information" is not supported by the evidence reviewed. The device generated meaningful activity on five channels, but: (a) the only genuinely human-authored AOSP SMS thread is self-directed; (b) the richest TextNow and WhatsApp exchanges either admit outright to generating test data ("Yep! Have to keep generating data") or are self-messaging between the device owner's own numbers/accounts; (c) three applications independently corroborate a single, consistent owner identity rather than suggesting a second,

unauthorised user. The evidence is more consistent with deliberate forensic-training-data generation than with covert, unauthorised communication - but this is a Tier 2 inference (Section 7), not a certainty, and the device's company-issued status and its allegation context mean genuine policy-relevant use cannot be fully excluded from this dataset alone.

Alternative interpretations

1. Test-environment creation (best-supported): the owner deliberately generated training data, as evidenced by two independent, explicit "generating data" / "switched over" statements and the overwhelmingly self-directed messaging pattern.

2. Compromised device or shared account: the WhatsApp registration number (919-579-4674) differs from the SIM number (919-579-0479), which can indicate SIM-swap or account sharing. Not supported by any other evidence of compromise (no unusual app installs, no root-compromise indicators) within this examination's scope.

3. Legitimate secondary-device use: the Pixel 3 was a secondary test/messaging device while the owner's primary number remained elsewhere, explaining the WhatsApp/TextNow/SIM number differences without implying unauthorised activity.

Additional evidence needed

- Carrier call-detail records for 919-579-0479 and 919-579-4674, to establish which physical SIM/device carried each number at each timestamp
- Wi-Fi connection logs (not just the residual SSID in WifiConfigStore.xml) to show where the device was actually associated
- locksettings.db, to determine lock type and whether biometric unlock was enrolled, and system logs (logcat/dmesg) for USB and unlock events
- Google account audit logs for thisisdfir@gmail.com, to show which devices accessed which services and when
- Chain-of-custody / physical-possession evidence external to this device image (witness statements, access records)

7. Attribution Assessment

Tier 1 - Direct observations

- Primary profile (user 0) display name "This Is"; aggregated AOSP contact "Josh Hickman"; Google account thisisdfir@gmail.com; TextNow account thisisdfir / "This Is DFIR" / thisisdfir@gmail.com; WhatsApp registration 919-579-4674.
- 24 SMS, 19 WhatsApp, 9 TextNow, 14 AOSP calls, 36 Google Messages rows, all directly queried and quoted in Section 5.

Tier 2 - Reasonable inferences

- The identity cluster above ("This Is" / "Josh Hickman" / thisisdfir) most likely represents a single individual, based on five converging, independent app-level sources.
- The device's messaging activity is best explained as deliberate test-data generation rather than covert personal or unauthorised communication, based on explicit statements in two separate applications and the predominantly self-directed message pattern.

Tier 3 - Limitations and alternative explanations

- Display names and usernames are self-entered and not biometrically or cryptographically verified.
- Multi-user device: a second profile (user 11, "User 2") exists; its activity was not examined in depth in this pass and is not attributed to user 0.
- No lock-screen/biometric evidence was captured, so physical-access control at any given time is unknown.
- Three distinct phone numbers are associated with the owner across apps (919-579-0479, 919-579-4674, 984-235-2054); the evidence does not establish why they differ.

Confidence levels

Claim	Confidence	Basis
Device is a Google Pixel 3 running Android 10, build QQ1A.200105.003	Very high	Direct build.prop properties, internally consistent across product and system partitions
Primary profile "This Is" is linked to identity cluster "Josh Hickman/thisisdfir"	High	Five independent, mutually corroborating app-level sources
A specific named individual physically operated the device at any given timestamp	Low / unresolved	No biometric, lock-state or external corroborating evidence available in this examination
Messaging activity was generated for test/training purposes rather than genuine unauthorised communication	Moderate-high	Explicit statements in TextNow and SMS content, self-directed pattern across three apps, but company-issued-device context not fully excluded
Device was physically present near Raleigh/Holly Springs, NC on 2019-09-17 and around 18:00-18:05 UTC on 2020-02-09	Moderate-high for the GPS-tagged points; low for exact duration/route	Direct GPS in photo EXIF and WhatsApp location shares; no continuous location trail

Defensible conclusion: the evidence supports that device profile "This Is" (user 0), linked with moderate-to-high confidence to the identity cluster "Josh Hickman/thisisdfir", generated the communication and location activity catalogued in this report, and that this activity is more consistent with deliberate test-data generation than with unauthorised third-party communication. It does not establish, to a forensic certainty, which physical person held the device at any specific moment, nor does it fully rule out the alternative interpretations in Section 6.

8. Conclusion

Across eight examination tasks, this report has established device identity and build provenance, two user profiles and a converging five-source identity cluster, settings and residual Wi-Fi configuration, a complete SMS/MMS/contacts/calls/calendar review, a Google-Messages/Dialer cross-validation exercise that surfaced RCS activity invisible to AOSP alone, WhatsApp and TextNow account and message reconstruction (including an explicit test-data admission), Maps and Photos location evidence with one genuine GPS cross-source corroboration, and a 21-event, six-family timeline with three supported correlations. The allegation of unauthorised communication and sharing of information is not supported by the evidence

reviewed and is better explained by deliberate test-dataset creation; attribution to a specific individual at a specific moment remains unresolved on this evidence alone.

9. Limitations and Issues Encountered During Analysis

Per the examination instructions, every tooling or methodology issue encountered is documented here together with how it was resolved or why it could not be.

I-01: Wrong build.prop property namespace (resolved)

The first grep against product/build.prop, using the conventional ro.product.* and ro.build.* key names, returned no output (Figure 5.1; Handling_Log row "Task 2.1a"). Android 10 introduced a separate /product/ partition, and its build.prop uses the partition-scoped namespaces ro.product.product.* (manufacturer/model/brand) and ro.product.build.* (build metadata) instead.

Fix: a diagnostic cat/grep -n pass over the full file identified the correct key names; the query was re-run using ro.product.product.* and succeeded (Section 5.1, Figure 5.2 for the companion system/build.prop query). The unprefixed ro.product.manufacturer seen at runtime on a live device is computed from these partition-prefixed values by init during boot and is not present as a literal key in the static file.

I-02: Device timezone initially misrecorded (resolved)

The case metadata initially recorded the analysis workstation's local timezone as EDT (UTC-04:00). A later cross-check of a database mtime against its stat offset showed the correct offset was -05:00 (EST), not -04:00.

Fix: corrected in the Handling Log ("Timezone correction" entry) before any UTC conversions were finalised for the report. This report's own timestamp conversions were performed independently from raw epoch values using UTC arithmetic throughout, so they are unaffected by this workstation-display issue either way; it is recorded here for completeness and because the assignment's own stated creator-documentation convention is UTC-05:00, consistent with the corrected value.

I-03: SMS _id/content mismatch between two raw-data captures (resolved)

Two independently captured, text-based sqlite3 query outputs (reports/raw_task4_1_telephony_sms.txt and the bugle_db cross-check in reports/raw_task4_5_bugle.txt) both labelled sms._id=12 as the message "Confirmed! To edit SMS preferences..." at 18:50:24 UTC. A direct, column-labelled DB Browser GUI screenshot of the same table (Figure 5.7) shows _id=12 is in fact "FB-55286 is ThisIs's Facebook confirmation code" at 18:50:13 UTC, and that _id=13 is the "Confirmed!..." message at 18:50:24 UTC.

Fix: the screenshot was treated as authoritative for the exact _id/content/time association, since it is a direct, unaltered, column-labelled capture of the live table that this examiner could visually verify, whereas the two CLI text captures agree with each other but not with the screenshot (apparent transcription/row-selection error carried from one capture into the other). This report (Section 5.3.2, Section 5.4) uses _id=13 for the "Confirmed!..." message throughout and flags the corresponding bugle_db cross-check label accordingly. This does not affect any count, hash or conclusion in the report - only the exact row-ID label for one message.

I-04: Unverified/incorrect UTC conversions in the supplementary analysis notes (resolved)

The supplementary notes supplied for this case (added_android.docx) included a full set of WhatsApp message times and one Google Maps saved-place time that do not match the UTC conversion of their own cited raw millisecond values - for example, WhatsApp message _id=12 was listed at 19:35:14 UTC against a raw value of 1581195314692 ms (which converts to 20:55:14 UTC), and the Maps saved-place entry was listed at 2019-04-05 15:49:03 UTC against a raw value of 1554407343604 ms (which converts to 2019-04-04 19:49:03 UTC). The notes themselves flagged this: "Some raw timestamps in my table were estimated from ms values; you should verify each against the actual DB values before final submission."

Fix: every timestamp used in this report (Sections 5.3-5.6, Section 6, and Timeline.csv) was independently recomputed in UTC directly from the raw epoch value captured in the reports/raw_*.txt evidence files, not taken from the supplementary notes' own arithmetic. Coordinates, message content, record IDs and counts in the supplementary notes were cross-checked

against the raw captures and found consistent except where separately noted (Issue I-03); only the clock-time conversions required correction.

I-05: Two recent camera photographs not resolved in the local_media query (unresolved - documented limitation)

A plain file listing of DCIM/Camera/ shows two recent photographs, IMG_20200209_132353.jpg and IMG_20200213_195030.jpg, that do not appear among the 8 rows returned by the local_media query used in this examination (Figure 5.9). The supplementary notes asserted specific capture times and an absence of GPS for both files, apparently derived from the Android camera filename convention (which encodes local capture time) rather than from a database row.

Resolution: not resolved within this examination's scope - re-running local_media without a LIMIT clause, or querying MediaProvider's own external.db, would likely surface these two rows and was identified but not completed in the time available. This report presents their timeline times explicitly labelled "filename-derived, unverified" rather than as confirmed database findings, and does not assert their GPS status one way or the other (Section 5.6, Issue I-05 cross-references).

I-06: Schema mismatches requiring probe-before-query (resolved)

Three queries failed on first attempt because the expected AOSP-style column name did not exist in that application's own schema: bugle_db.participants (expected phone_number; column does not exist in this build), msgstore.messages (expected chat_row_id; the actual column is key_remote_jid/key_from_me), and local_media (expected is_deleted; this column does not exist in the Photos app's local_media table, only in its cloud-synced media table).

Fix: each failure was followed by a PRAGMA table_info() schema probe against the actual table before re-running the query with the correct column names (see Queries.sql, Task 6a and Task 7b sections, for both the corrected queries and an inline note on the probe). This is standard practice when an app's internal schema is undocumented and differs from the AOSP reference schema the task brief assumes, and all three corrected queries subsequently returned usable data, reflected throughout Sections 5.4-5.6.

I-07: Binary/garbled characters in Google Dialer text export (documented, no forensic impact)

The AnnotatedCallLog export (reports/raw_task5_dialer.txt) contains non-printable/garbled byte sequences interleaved with several rows' number/country fields when read as plain text.

Assessment: this is consistent with a BLOB-typed column (e.g. an internal lookup key or icon reference) being included in a SELECT * style export and rendered as text; the numeric, time and duration fields in the same rows are intact and were cross-validated against AOSP calllog.db (Section 5.4), so no finding in this report depends on the corrupted field. Future extraction should explicitly list only the needed columns to avoid this.

Appendix A: Evidence Register

Full register with source paths, sizes, SHA-256 hashes and capture times is provided as the accompanying Evidence_Register.csv. Key entries:

Evidence ID	Description	SHA-256 (abbreviated)
E-004	data/system/users/0.xml	67bfa3b7...ccb3adb
E-010	mmssms.db (CE, authoritative)	cd5452f5...53f1342
E-011	mmssms.db (DE, corroboration)	d0ce9034...34c827
E-014	contacts2.db	2a1fee3d...4625050
E-015	calllog.db	5979a3fa...ef809d3
E-017	calendar.db	82a58090...039e5aca3c
E-018	bugle_db	ed4f19eb...57ea7e3a33
E-019	annotated_call_log.db	e7b896b1...6f6a417e59
E-023	WhatsApp msgstore.db	9e133d72...496dd13105
E-025	Google Photos gphotos-1.db	55caad23...449de85a8331e20
E-026	TextNow textnow_data.db	3ceb390e...aeddc75edc86

Appendix B: Screenshot Reference

Figure	File	Content
3.1	tools.png	Environment and tool-version verification
3.2	pixel3_file.png	Evidence archive located
5.1	product_build.png	Failed product/build.prop grep (Issue I-01)
5.2	system_build.png	system/build.prop properties
5.3	line_count.png	Line-count verification of both build.prop files
5.4	users.png	data/system/users/ directory listing
5.5	userlist.png	userlist.xml
5.6	owner.png	0.xml primary user profile
5.7	SMS.png	mmssms.db sms table (DB Browser) - resolves Issue I-03
5.8	media_table.png	Google Photos media table
5.9	local_media.png	local_media table and gmm_storage_table BLOB previews
(ref)	working_dir.png	Case directory structure created at acquisition
(ref)	working_pixel3_file.png	Extracted Pixel 3 working folder

Appendix C: Timeline (12+ events, see Timeline.csv for full detail)

#	UTC time	Event	Evidence ID	Family
1	2019-12-17 19:34:12	ROM compiled (QQ1A.200105.003)	E-001/E-002	Build properties
2	2019-04-04 19:49:03	Maps saved place near Raleigh, NC	E-024	Maps
3	2019-09-17 08:26:37	GPS-tagged photo captured, ~100m from Maps entry	E-025	Photos
4	2020-01-29 18:20:45	First SMS received (LINE code)	E-010	SMS
5-6	2020-01-29 18:50:13 / 18:50:24	FB-55286 code, then Thread 13 opens	E-010	SMS
7-8	2020-02-08 20:55:14 / 20:57:49	WhatsApp conversation with self-contact begins	E-023	WhatsApp
9-10	2020-02-08 20:00:46 / 20:03:06	TextNow conversation incl. "generating data" admission	E-026	TextNow

#	UTC time	Event	Evidence ID	Family
11-12	2020-02-09 18:00:14 / 18:05:02	WhatsApp live + static location shared	E-023	WhatsApp
13	2020-02-09 18:23:53 (unverified)	Camera photo captured (filename-derived)	E-027	Photos
14	2020-02-09 18:41:42	SMS "coffee shop" sent	E-010	SMS
15	2020-02-12 15:37:29	Missed call + voicemail	E-015	Calls
16-19	2020-02-14 01:29-14:41	Four self-calls	E-015	Calls
20	2020-02-14 15:00:54	Last write to CE mmssms.db	E-010	Filesystem
21	2020-02-14 15:51:57	User 0 last login	E-004	User profile

Appendix D: Handling Log Summary

Full handling log (42 rows) is provided as the accompanying Handling_Log.csv, documenting every acquisition, hashing, and query action from case-directory creation through Task 5, including the two "FAILED" entries (Task 2.1a build.prop namespace, logged and resolved per Issue I-01 above) and the explicit timezone correction (Issue I-02).